Generative Modeling

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The Two Philosophies of Machine Learning

The Geometric School (Empirical Risk Minimization):

- We do not care how the universe generated the data. We only care about drawing boundaries or surfaces that minimize a penalty.
- *Language:* Distances, margins, hyperplanes, gradients.
- *Examples:* Support Vector Machines, Decision Trees, standard Logistic Regression.

The Statistical School (Probabilistic Modeling):

- We assume data is the output of a hidden random process. We model the underlying probability distributions and find the parameters that make our observations most likely.
- *Language:* Maximum Likelihood, priors, variance, Bayes' Theorem.
- *Examples:* Naive Bayes, Discriminant Analysis.

The Two Philosophies of Machine Learning

- **The Revelation:** We have actually used both schools already! Let's look at Linear Regression.
- **Geometric view:** Minimize the sum of squared distances (Least Squares).
- **Statistical view:** Assume the data has a deterministic trend plus Gaussian noise ($\varepsilon \sim N(0, \sigma^2)$). If we use Maximum Likelihood Estimation (MLE) to find the best fit...
- **The Math:** The negative log-likelihood of a Gaussian cancels out the exponential, dropping the constants, leaving us exactly with the goal of minimizing $\Sigma(y - y\text{^})^2$
- *Takeaway:* Assuming Gaussian noise in statistics is mathematically identical to applying a squared-error penalty in geometry!

The Two Philosophies of Machine Learning

Why did we start with the Geometric approach?

- **Reason 1: Vapnik's Principle.**
  - *Vladimir Vapnik (co-inventor of the SVM) famously said: "When solving a problem of interest, do not solve a more general problem as an intermediate step."*
  - Estimating the full joint distribution of the universe P(x, y) is a vastly harder mathematical problem than just drawing a boundary P(y|x). For classification, generating the data is an unnecessary intermediate step. We started with the easiest, most direct path to the solution.
- **Reason 2: The Deep Learning Bridge.**
  - The entire modern deep learning revolution is built on empirical risk minimization. By spending the first portion optimizing loss landscapes and calculating gradients, you now possess the exact mathematical toolkit required to understand Neural Networks.
- **Reason 3: The Curse of Dimensionality.**
  - Generative models are incredibly fragile in high dimensions unless you make massive simplifying assumptions. Geometric models like SVMs (with the Kernel trick) and Ensembles handle infinite-dimensional spaces natively without breaking a sweat. We taught you the tools that scale to big data first.

Generative and Discriminative Classifiers

- **Generative Classifiers**
  A generative classifier tries to learn the model that generates the data behind the scenes by estimating the assumptions and distributions of the model. It then uses this to predict unseen data, because it assumes the model that was learned captures the real model. As we will see, this is often not true.

- **Discriminative Classifiers**
  A discriminative classifier tries to model by just depending on the observed data. It makes fewer assumptions on the distributions but depends heavily on the quality of the data (Is it representative? Are there a lot of data?).

Generative Classifiers

- Generative methods focus on modeling P(X, G) and then use G as a classifier where

$$\begin{aligned}\hat{G}(\mathbf{X}) \;:=\; \underset{G}{\operatorname{argmax}}\, \hat{\mathsf{P}}(G|\mathbf{X}) &= \underset{G}{\operatorname{argmax}} \frac{\hat{\mathsf{P}}(\mathbf{X}|G)\hat{\mathsf{P}}(G)}{\sum_i \hat{\mathsf{P}}(\mathbf{X}|G_i)\hat{\mathsf{P}}(G_i)} \\ &= \underset{G}{\operatorname{argmax}}\, \hat{\mathsf{P}}(\mathbf{X}|G)\hat{\mathsf{P}}(G)\end{aligned}$$

- Examples include: Naive Bayes, Linear Discriminant Analysis (LDA), Quadratic Discriminant Analysis (QDA)

Gaussian Discriminant Analysis

- **The Core Assumption:** To make generative models work for continuous data, we must assume the features follow a Multivariate Normal distribution: $P(x|y=k) \sim N(\mu_K, \Sigma_k)$.
- **The GDA Family Tree:** Everything we learn today is just a variation of how strict we want to be about the Covariance Matrix (Σ).
  - **QDA (Quadratic):** Every class gets its own unique covariance matrix. Highly flexible, highly prone to overfitting.
  - **LDA (Linear):** All classes share the exact same global covariance matrix. Cancels out quadratic math, highly robust.
  - **Naive Bayes (Diagonal):** We force all off-diagonal covariance terms to 0. Assumes perfect feature independence.
  - **RDA (Regularized):** A blend of QDA and LDA using a shrinkage penalty.

Naive Bayes

- Naive Bayes is a generative classifier that estimates P(**X**, G) by assuming

$$\mathsf{P}(\mathbf{X}|G) = \prod_j \mathsf{P}(X_j|G)$$

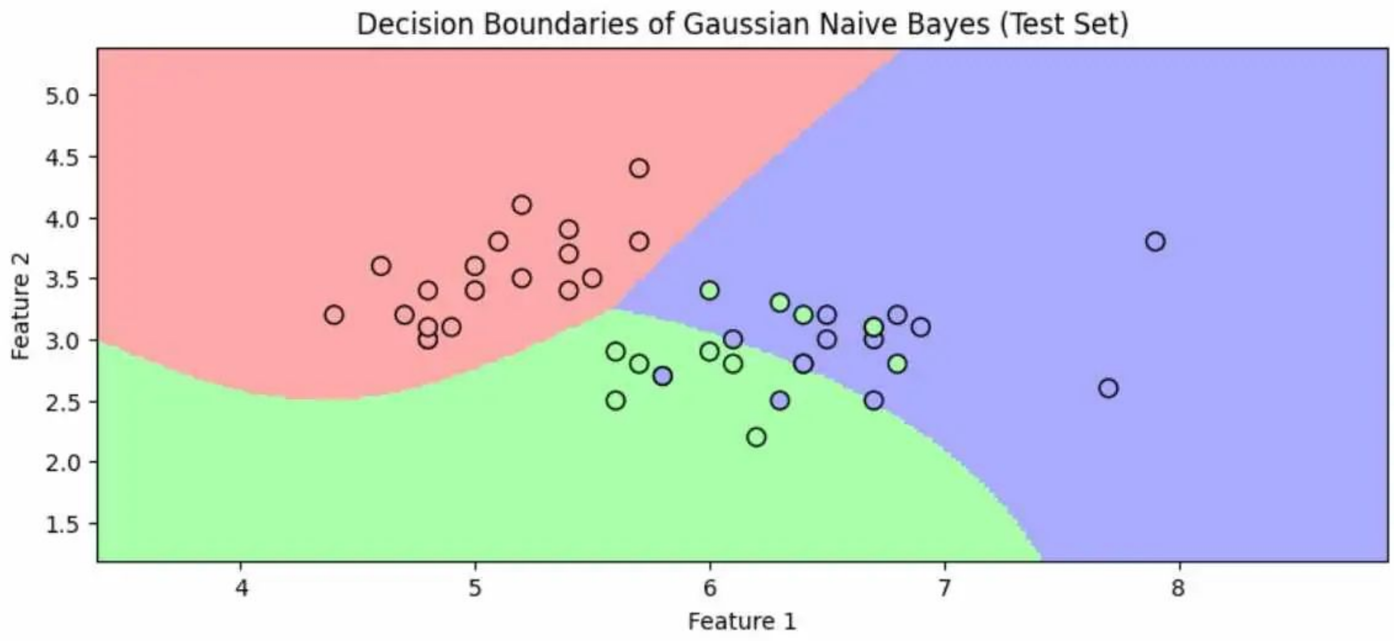
- Thus, the features are independent conditional on G.
- Since P(**X**, G) = P(**X**|G) P(G), naive Bayes estimates P(G) and the P($\mathbf{X}_j$ |G) for all j separately via MLE and then classifies according to

$$\begin{aligned}
\hat{G}(\mathbf{x}) &= \underset{G \in \mathcal{G}}{\operatorname{argmax}}\, \hat{\mathsf{P}}(G|\mathbf{X} = \mathbf{x}) \\
&= \underset{G \in \mathcal{G}}{\operatorname{argmax}}\, \hat{\mathsf{P}}(G)\hat{\mathsf{P}}(\mathbf{x}|G) \\
&= \underset{G \in \mathcal{G}}{\operatorname{argmax}}\, \hat{\mathsf{P}}(G) \prod_i \hat{\mathsf{P}}(x_i|G)
\end{aligned}$$

Naive Bayes

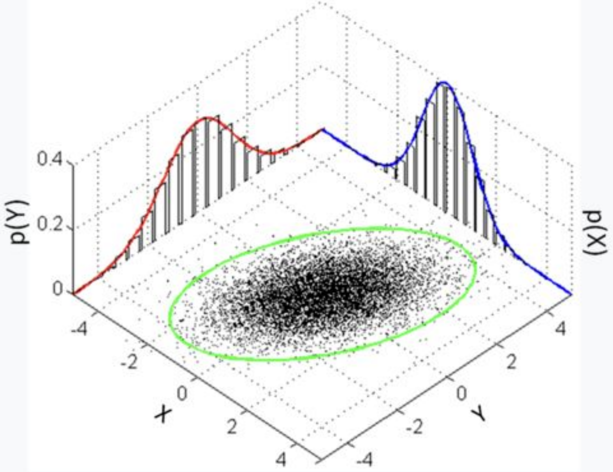
- **The Curse of Dimensionality:** Why do we assume $P(X|G) = \Pi\ P(X_j|G)$? Because calculating a joint distribution for 1,000 features requires estimating a massive number of parameters.
- **The GDA Connection:** Think of the Family Tree. By assuming features are independent, we are mathematically zeroing out all the off-diagonal correlation terms in our covariance matrix.
- **Laplace Smoothing:** If a word never appears in a spam email in the training set, its probability is 0. Because of the product rule, one zero destroys the whole equation. We fix this by adding a small constant (α) to all counts.

Naive Bayes


Decision Boundaries of Gaussian Naive Bayes (Test Set)
Feature 2
5.0
4.5
4.0
3.5
3.0
2.5
2.0
1.5
4
5
6
7
8
Feature 1

Multivariate Normal Distribution

PDF	$(2\pi)^{-k/2}\det(\boldsymbol{\Sigma})^{-1/2}\,\exp\left(-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^{\mathrm{T}}\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\right),$ exists only when $\boldsymbol{\Sigma}$ is positive-definite



Many sample points from a multivariate normal distribution with $\boldsymbol{\mu} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ and $\boldsymbol{\Sigma} = \begin{bmatrix} 1 & 3/5 \\ 3/5 & 2 \end{bmatrix}$, shown along with the 3-sigma ellipse, the two marginal distributions, and the two 1-d histograms.

Linear Discriminant Analysis (LDA)

- LDA is a generative model that assumes class-conditional densities, $f_k(x)$, are Gaussian with a common covariance matrix so that

$$f_k(\mathbf{x}) \;=\; \frac{1}{(2\pi)^{M/2}|\boldsymbol{\Sigma}|^{1/2}} e^{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu}_k)^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu}_k)}$$

$$\hat{G}(\mathbf{x}) = \operatorname*{argmax}_k \frac{\hat{f}_k(\mathbf{x})\hat{\pi}_k}{\sum_{j=1}^K \hat{f}_j(\mathbf{x})\hat{\pi}_j}$$

$$\begin{aligned} \hat{\pi}_k &= N_k/N \\ \hat{\boldsymbol{\mu}}_k &= \sum_{g_i=k} \mathbf{x}_i/N_k \\ \hat{\boldsymbol{\Sigma}} &= \frac{\sum_{k=1}^K \sum_{g_i=k} (\mathbf{x}_i - \hat{\boldsymbol{\mu}}_k)(\mathbf{x}_i - \hat{\boldsymbol{\mu}}_k)^\top}{N-K} \end{aligned}$$

Linear Discriminant Analysis (LDA)

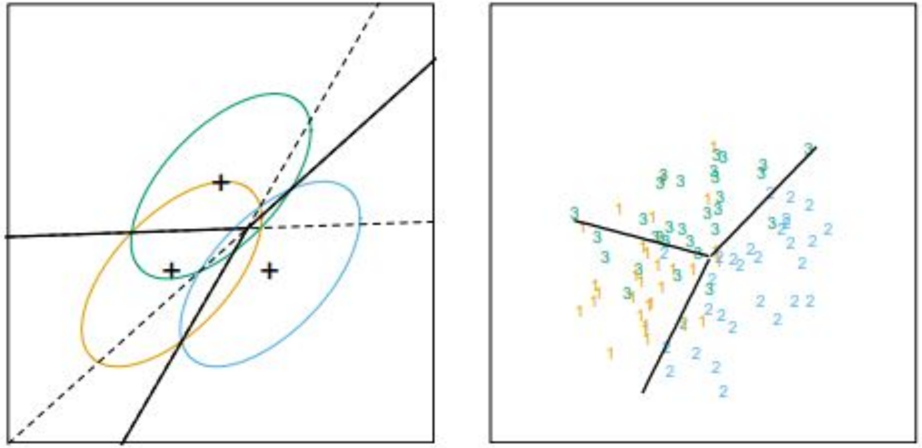


FIGURE 4.5. *The left panel shows three Gaussian distributions, with the same covariance and different means. Included are the contours of constant density enclosing 95% of the probability in each case. The Bayes decision boundaries between each pair of classes are shown (broken straight lines), and the Bayes decision boundaries separating all three classes are the thicker solid lines (a subset of the former). On the right we see a sample of 30 drawn from each Gaussian distribution, and the fitted LDA decision boundaries.*

Linear Discriminant Analysis (LDA)

- log-ratio between two classes, k and l, to get

$$\log \frac{\mathsf{P}(G=k|\mathbf{X}=\mathbf{x})}{\mathsf{P}(G=l|\mathbf{X}=\mathbf{x})} = \log \frac{\hat{\pi}_k}{\hat{\pi}_l} - \frac{1}{2}\left(\hat{\boldsymbol{\mu}}_k + \hat{\boldsymbol{\mu}}_l\right)^\top \hat{\boldsymbol{\Sigma}}^{-1}\left(\hat{\boldsymbol{\mu}}_k - \hat{\boldsymbol{\mu}}_l\right) + \mathbf{x}^\top \hat{\boldsymbol{\Sigma}}^{-1}\left(\hat{\boldsymbol{\mu}}_k - \hat{\boldsymbol{\mu}}_l\right)$$

- We define linear discriminant functions:

$$\delta_k(\mathbf{x}) := \mathbf{x}^\top \hat{\boldsymbol{\Sigma}}^{-1} \hat{\boldsymbol{\mu}}_k + \underbrace{\log \hat{\pi}_k - \frac{1}{2}\hat{\boldsymbol{\mu}}_k^\top \hat{\boldsymbol{\Sigma}}^{-1} \hat{\boldsymbol{\mu}}_k}_{k^{th}\ \text{intercept}}$$

- The LDA estimator is:

$$\hat{G}(\mathbf{x}) = \underset{k}{\operatorname{argmax}}\, \delta_k(\mathbf{x})$$

Quadratic Discriminant Analysis (QDA)

- A generalization to linear discriminant analysis is quadratic discriminant analysis (QDA).
- Why do you suppose the choice in name?
- The implementation is just a slight variation on LDA. Instead of assuming the covariances of the MVN distributions within classes are equal, we instead allow them to be different.

Quadratic Discriminant Analysis (QDA)

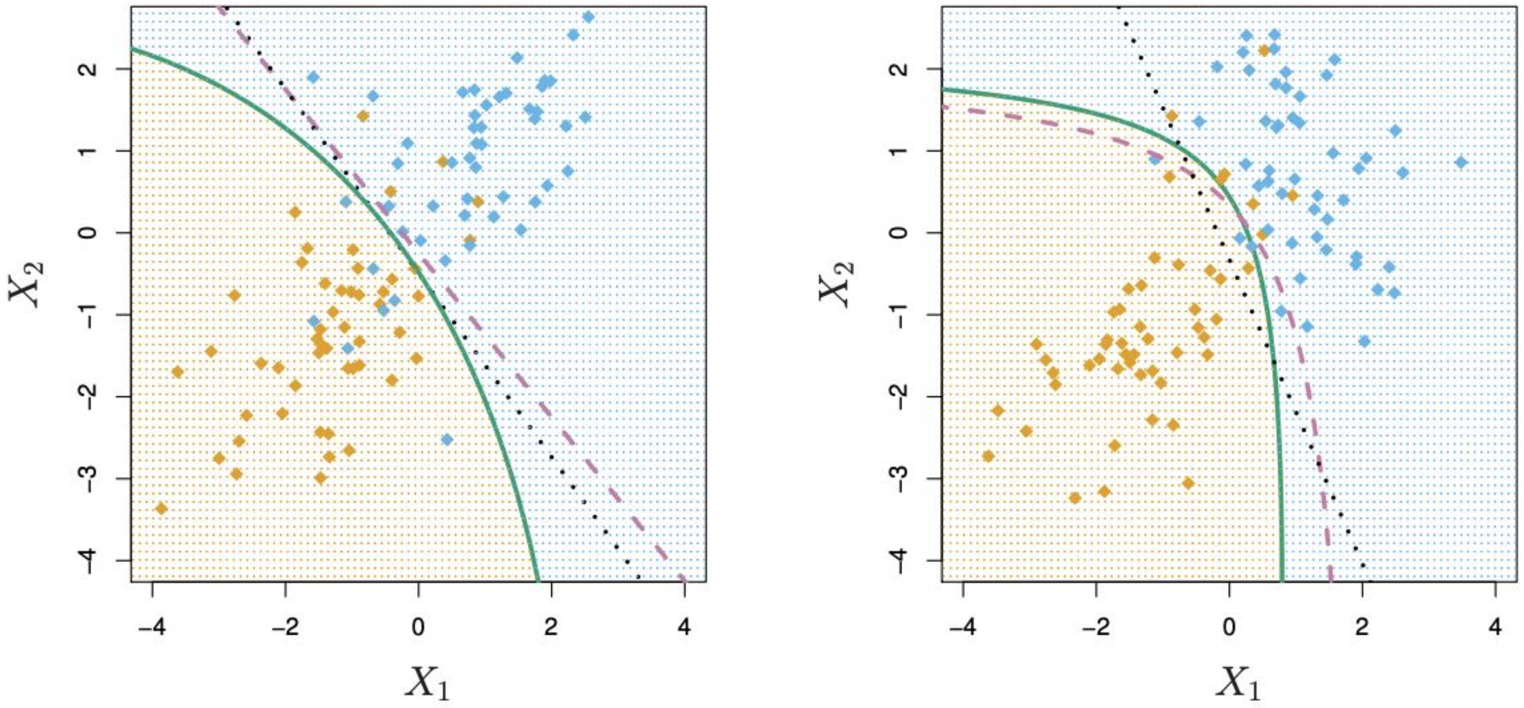
- Drop equal covariance assumption and obtain quadratic discriminant functions

$$\delta_k(\mathbf{x}) \;:=\; -\frac{1}{2}\log|\hat{\boldsymbol{\Sigma}}_k| \;-\; -\frac{1}{2}(\mathbf{x}-\hat{\boldsymbol{\mu}}_k)^\top \hat{\boldsymbol{\Sigma}}_k^{-1}(\mathbf{x}-\hat{\boldsymbol{\mu}}_k) \;+\; \log\hat{\pi}_k$$

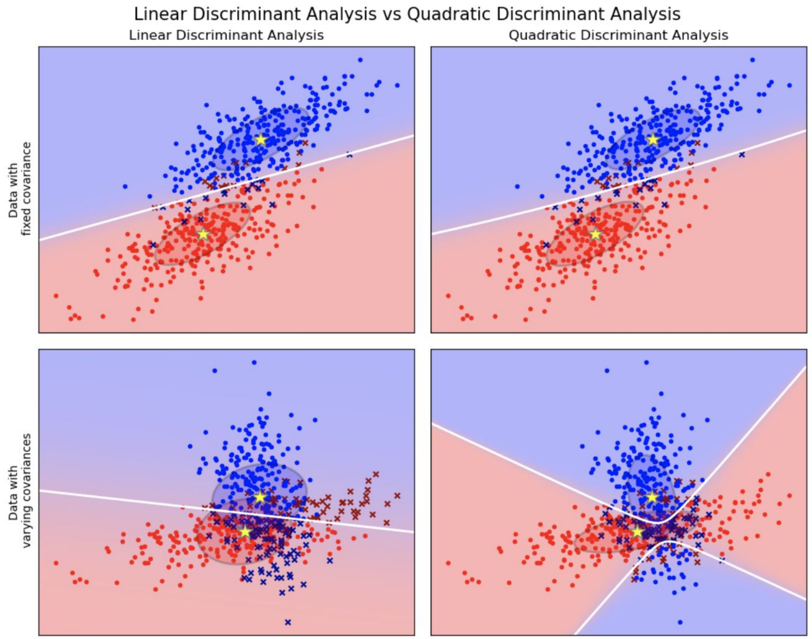
- The QDA classifier is

$$\hat{G}(\mathbf{x}) = \underset{k}{\operatorname{argmax}}\, \delta_k(\mathbf{x})$$

LDA vs QDA



LDA vs QDA



Linear Discriminant Analysis vs Quadratic Discriminant Analysis
Linear Discriminant Analysis
Quadratic Discriminant Analysis
Data with fixed covariance
Data with varying covariances

Regularized Discriminant Analysis (RDA)

- **The Problem with LDA (High Bias):** By forcing all classes to share the exact same covariance matrix Σ, LDA is highly rigid. It is incredibly stable on small datasets, but it will massively underfit if the classes actually have different shapes.
- **The Problem with QDA (High Variance):** By giving every class its own Σ_k, QDA is highly flexible. But a covariance matrix for p features has p(p+1)/2 parameters. If you have 50 features and 10 classes, QDA requires estimating over 12,000 parameters!
- **The Breaking Point:** If a class has fewer training examples than the number of features, calculating the inverse matrix Σ^{-1} becomes mathematically impossible (the matrix is singular). QDA completely breaks down.
- **The Question:** Can we find a sweet spot between the rigidity of LDA and the fragility of QDA?

Regularized Discriminant Analysis (RDA)

- **The Intuition:** Just like we used a penalty term in Ridge Regression to "shrink" extreme weights, we can "shrink" the individual QDA covariance matrices toward the shared LDA covariance matrix.
- **The Mathematics of Shrinkage:** RDA introduces a hyperparameter α (where 0 <= α <= 1) to compute a new, blended covariance matrix for each class:

$$\hat{\Sigma}_k(\alpha) = \alpha\hat{\Sigma}_k + (1-\alpha)\hat{\Sigma}$$

  (Where Σ^$_k$ is the QDA matrix, and Σ^ is the pooled LDA matrix).
- **How it works:** We evaluate the discriminant function $\delta_k(x)$ using this new blended matrix Σ^k(α) instead of the extremes.

Regularized Discriminant Analysis (RDA)

The Extremes:

- If $\alpha = 0$, the model is exactly **LDA**.
- If $\alpha = 1$, the model is exactly **QDA**.

The Middle Ground: Any value between 0 and 1 creates a boundary that is curved (like QDA) but constrained and smoothed out (like LDA).

How do we choose α? Like any hyperparameter, it cannot be learned directly from the training data (setting it on training data would always push α -> 1 to overfit). We must find the optimal α using **Cross-Validation** on a validation set.

Further Regularization: In highly extreme cases (thousands of features), RDA can apply a *second* shrinkage parameter, γ, which shrinks the pooled matrix toward a purely diagonal matrix (essentially shrinking the model all the way down to Naive Bayes!

GDA vs Logistic Regression

The Paradox: LDA gives a linear boundary ($w^Tx + b = 0$). Logistic Regression gives a linear boundary ($w^Tx + b = 0$). Are they the same?

The Answer: No. They optimize different things.

- **LDA (Generative):** Assumes the data is Gaussian. If the data *is* Gaussian, LDA finds the perfect boundary faster (needs less data) than Logistic Regression.
- **Logistic Regression (Discriminative):** Makes no assumptions about the input data distribution. If the data is skewed, messy, or has extreme outliers, Logistic Regression will wildly outperform LDA.